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To cite this article: Sijia Wang, Jun Chen, Dongyang Hou, Xiaoguang Zhou, Bingliang Cui, Xiaoyang Zhang, Yu Wang, Dongyuan Li, Junyu Cui, Silong Luo & Qiankun Kang (2025) Development and application of knowledge graph-based spatiotemporal data model for urban physical examination, International Journal of Digital Earth, 18:1, 2512060, DOI: [10.1080/17538947.2025.2512060](https://doi.org/10.1080/17538947.2025.2512060)

To link to this article: <https://doi.org/10.1080/17538947.2025.2512060>



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Published online: 03 Jun 2025.



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Development and application of knowledge graph-based spatiotemporal data model for urban physical examination

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ABSTRACT

The urban physical examination is pivotal in diagnosing and resolving 'urban diseases'. However, it encounters challenges including the intricate interconnections among heterogeneous data, the spatiotemporal differences in urban pathologies, and the multi-dimensional scenario-oriented representation. A knowledge graph is a potent technical instrument capable of depicting data replete with relational details and capturing the dependency ties among entities. In view of the above, this paper proposes a knowledge graph-based spatiotemporal data model for the urban physical examination. The model focuses on application scenarios, and builds a conceptual model with a three-layer architecture of 'semantics-data-scenario' and an ontology logic model structured as a hypergraph. This is intended to facilitate the management of spatiotemporal data throughout the process, while also enabling scenario-based spatiotemporal expression. Furthermore, this paper presents an in-depth analysis of the urban greenway construction case. The results show that the model organizes and expresses urban data and knowledge, covering multiple levels (such as themes, connotations, knowledge points, and indicators) and multiple granularities (macro-city, meso-region, micro-street), helping to understand urban physical examination from multiple dimensions. This not only provides a scientific basis for urban planners to make decisions but also sets an example for the practice of urban physical examination knowledge service.

ARTICLE HISTORY

Received 29 December 2024

Accepted 16 May 2025

KEYWORDS

Urban physical examination; knowledge graph; knowledge service; spatiotemporal data model; ontology

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1. Introduction

The principal objective of urban physical examination, alternatively termed urban health assessments, is to quantify and analyze the city's spatial configuration, demographic distribution, resource allocation, and environmental quality. These factors are collectively referred to as 'vital signs' and are used to ascertain the health status and developmental trajectories of urban systems (Barton and Grant 2013; Friel et al. 2011; Galea and Vlahov 2005; Harpham 2009; Vardoulakis, Dear, and Wilkinson 2016). This process is data-driven, relying on a spectrum of quantified indicators that reflect these urban vital signs for the monitoring, diagnosis, early warning, and maintenance of urban development (Allen et al. 2017; Andries et al. 2019; Pineo et al. 2018).

Internationally, urban physical examination predominantly entails the establishment of a comprehensive evaluation indicator system of urban physical signs (Corburn 2004; Sharifi and Murayama 2014). Specifically, by collecting data on urban infrastructure, public services, social economy, ecological environment, cultural features and other aspects, combined with empirical materials such as questionnaires and social feedback, and applying methods like structural decomposition, inventory statistics, model calculation, standard setting, health status classification, and comprehensive measurement, a multi-dimensional comparative analysis and evaluation are conducted on all aspects of the urban 'living organism' (Biswas et al. 2021; Dempsey et al. 2011; He et al. 2018). Since the 1990s, the international scientific community has consistently placed significant emphasis on pre-implementation appraisal, process monitoring, and post-implementation evaluation within the scene of urban development planning (Naess 2001; Yigitcankar and Teriman 2015). It not only gauges the degree of congruence between the implementation outcomes and the planning objectives but also accords greater attention to sundry urban health issues that impinge upon the implementation process. The United Kingdom, Japan and the United States have all formulated strategic planning objectives, constructed quantifiable indicators and thus formed the core content of monitoring and evaluation (Sharifi and Murayama 2014). Then, centering around these indicators, they have aggregated multi-source spatio-temporal data that are dynamic and timely to build a monitoring and evaluation database. Eventually, by combining quantitative and qualitative indicator evaluation methods, they have provided strong support for the implementation of planning and the action plans of local governments (Geneletti et al. 2017). Nevertheless, this process also faces the challenges of width of data sources, the depth of indicator diagnosis, and the efficacy of comprehensive data application (Childers et al. 2014). In February 2017, the Chinese government proposed the establishment of an assessment mechanism for urban physical examination, and since then many cities in China have carried out related work (Zhang, He, and Chen 2021). Among them, as one of the assessment items of urban physical examination, urban greenways can promote green travel and reduce carbon emissions, so their construction and development have received much attention (Semeraro et al. 2021; Van Oijstaeijen, Van Passel, and Cools 2020). In general, urban physical examination rely on strategic planning orientation, evaluation index, and spatiotemporal data collection (Cui et al. 2024; Ren et al. 2024). However, the strategic planning objectives of urban physical examination are different, which makes it difficult to make horizontal comparisons of assessment indicators and carry out multi-level vertical analyses between 'objectives-indicators-data'

(He et al. 2023). Furthermore, in practice, too much attention is paid to the urban physical examination indicators themselves, ignoring the semantic association between the indicators and data, which in turn makes it difficult to explore the intrinsic connection and potential value of urban physical signs (Chen et al. 2022). In addition, the current urban physical examination lacks a spatiotemporal data model based on application scenarios, which makes it difficult to reflect its spatiotemporal characteristics and changes (Cook and Karvonen 2024). Therefore, high-quality urban physical examination not only needs to consider the complex interactions between planning objectives and assessment indicators, but also needs to address the differences in dimensionality and granularity of different spatiotemporal data.

In recent years, there has been a gradual shift in the urban physical examination methodology from the utilization of socio-economic statistical data to the incorporation of multi-source spatial and temporal data, such as remotely sensed imagery, points of interest, human tracks, and street view images (Giles-Corti, Lowe, and Arundel 2020; Masó et al. 2019; Stokes and Seto 2019). It is not sufficient to store these voluminous multidimensional data sets, but it is also imperative to guarantee the correlation and semantic consistency between the data, to facilitate the intricate analysis and decision-making processes. However, current research mostly relies on traditional relational or non-relational databases, such as files, key-value pairs, attribute tables, etc., which are only suitable for dealing with structured data and simple relational queries, and lack the organization and management of spatiotemporal data (Codd 1970). And the current spatiotemporal data model of urban physical examination has not yet been developed, which makes it difficult to represent highly interconnected data and complex semantic relationship networks. This restricts the deep integration and analysis capabilities at the level of urban physical examination spatiotemporal data, and it is difficult to meet the data organization and management needs. It is mainly reflected in the following three aspects: (1) Variable Application Scenarios: The urban physical examination encompasses multiple stakeholders and variables, presenting challenges in designing and executing effective protocols (Wirth et al. 2014); (2) Insufficient Knowledge Articulation: The assessments involve knowledge covering numerous themes, connotations, knowledge points, and indicators. This renders the expression, understanding, and visualization of such knowledge a complex task (Sajjad, Chan, and Chopra 2021); (3) Intricate Data Associations: The data dealt with in assessments have spatiotemporal and semantic linkages and are characterized by high-dimensional dynamics. This complicates tasks like data analysis, mining, reasoning, updating, and traceability (Hamdi et al. 2022). Consequently, the effective integration of the semantic and logical representation of urban vital signs, the management of the entire lifecycle of spatiotemporal data, and the provision of scenario-specific presentations to amplify the application efficacy of data and knowledge have emerged as pivotal issues in contemporary research.

The Knowledge Graph (KG), an advanced information organization technology, utilizes graph databases to interconnect multi-source data via entities (nodes) and relationships (edges), thereby constructing a unified data view conducive to complex data analysis and knowledge discovery (Abu-Salih 2021; Ji et al. 2021). It links dispersed data points into a network, which not only facilitates comprehensive assessment of urban vital signs but also uncovers implicit knowledge from massive data through reasoning and analysis, providing deeper insights and decision-making support

(Chadzynski et al. 2021; Janowicz et al. 2022). Preliminary progress has been achieved in the application of KG for to urban spatiotemporal data. For example, WorldKG has linked OpenStreetMap's geospatial data with knowledge from various fields by aligning with other KGs like Wikidata and DBpedia, enhancing data interoperability (Dsouza et al. 2021). The UK's National Digital Twin (NDT) unifies disparate data sources through a semantic model, integrating real-time data streams and computational agents to form a knowledge network for supporting sophisticated analyses (Akroyd et al. 2021). Moreover, the CityGML-KG framework transforms 3D urban model data into a KG, integrating with sources such as OpenStreetMap and enhancing data semantic expression (Ding et al. 2024). Domestically, researchers are actively applying KG in urban planning and management. For instance, the UrbanKG system combines street view imagery, satellite imagery, and user behavior data to construct a knowledge graph that facilitates the integration of multimodal datasets, encompassing text, images, and sequences (Liu et al. 2023). UrbanGPT integrates large language models (LLMs) into spatiotemporal data analysis, enabling the model to handle not only natural language but also understand and predict complex spatiotemporal dynamics (Li et al. 2024). Furthermore, some scholars have introduced KG technology into urban physical examination and developed KG-driven collaborative methods for multi-inspection tasks, realizing the synergy of 'data-indicator-calculation' among multiple types of urban physical examinations (Wang et al. 2022). This has brought new tools and demonstrated great application potential. For example, the utilization of RDF triple stores and native graph database storage, such as Jena, RDF4J, Neo4j, etc., has enhanced the integration and analysis capabilities at the data level (Fernandes and Bernardino 2018; Wang et al. 2019). It should be noted that although current research can support RDF graphs and attribute maps to achieve spatiotemporal data management based on knowledge graphs, however, these studies are still insufficient to support the demand for in-depth integration and analysis applications of urban physical signs (Chen, Liu, and Tian 2022; Krishnan, Dziedzic, and Elmore 2018; Liu et al. 2021a; 2021b). Specifically, when dealing with spatiotemporal data, the integration of multi-source data is often incomplete, resulting in data fragmentation and making it difficult to comprehensively and accurately reflect the complex situation of urban physical examination. In terms of knowledge mining, most studies fail to construct an effective conceptual model to deeply explore the potential relationships among data, leading to a lack of in-depth scientific basis for planning decisions (Akroyd et al. 2021). Additionally, with respect to result presentation and visualization, current studies have difficulty in intuitively and effectively conveying key information to urban planners and decision-makers, which is not conducive to multi-departmental collaboration and public participation (Chen et al. 2023).

In view of the above, this paper proposes a knowledge graph-based spatiotemporal data model for the urban physical examination. Initiating from application scenarios, it aims to integrate multi-source heterogeneous spatiotemporal data more effectively through knowledge graph and explore the complex relationships among spatiotemporal data, to optimize the organization and management of spatiotemporal data for urban physical examination knowledge services. The model adopts a three-layer architecture of 'semantics-data-scenario', which constructs the urban physical examination knowledge system, manages the whole process spatiotemporal data of urban physical examination and provides scenario-based spatiotemporal expression. This paper presents the

conceptual and logical models of the proposed model, and provides relevant example and explanations by taking the urban greenway construction as an instance.

- (1) Currently, most urban physical examination spatiotemporal data models concentrate on the design of new indicators and conceptual discussions. Although progress has been made in the organization and association of data and semantics in urban physical examination knowledge graphs, the requirements of urban physical examination application scenarios have not been fully considered. This paper proposes a knowledge graph-based spatiotemporal data model for urban physical examination. The model integrates the knowledge in the semantic layer and the data in the data layer according to the requirements of application scenarios. Then, it presents them in an intuitive and highly interactive visual manner through the scenario layer to obtain customized knowledge services.
- (2) Design a logical model integrating UML diagrams and hypergraph ontology. This logical model represents high-order relations as hyperedges, simplifies the structure of UML models, and enhances the accuracy of expressing multi-dimensional relations.
- (3) Conduct model validation with the example of Guangzhou greenway construction. Based on the model, the example identifies and diagnoses the problems in the construction of Guangzhou greenways, puts forward targeted strategies, and improves decision-making efficiency by means of visualization, thus providing a visual knowledge service example for the management of urban greenway construction.

2. General framework

Within the urban physical examination spatiotemporal data management, conventional Entity-Relationship (ER) modeling approaches encounter limitations in scalability. They struggle to capture the interconnections between urban physical examination spatiotemporal data and associated knowledge domains or application scenarios (Malinowski and Zimányi 2008; Sharifi 2021; Teorey, Yang, and Fry 1986; Zhou et al. 2021). Thus, to enhance the efficient organization and management of urban physical examination spatiotemporal data, this study introduces a knowledge graph-based spatiotemporal data model for urban physical examination, as shown in Figure 1. Starting from application scenarios, this model builds a comprehensive urban physical examination knowledge framework, manages the entire spatiotemporal data lifecycle, and provides scenario-specific spatiotemporal expressions. It integrates various spatiotemporal data types and relational structures into the unified framework.

Firstly, this paper proposes a three-layer framework conceptual model of ‘semantics-data-scenario’ around the application scenario requirements. Within this framework, the semantics layer abstracts and defines the spatiotemporal semantics of urban vital signs. The data layer is responsible for integrating and managing the spatiotemporal data across the whole process. The scenario layer enables the conversion of abstract data and knowledge into perceptible and visual representations of urban development narratives via scenario-specific spatiotemporal presentations.

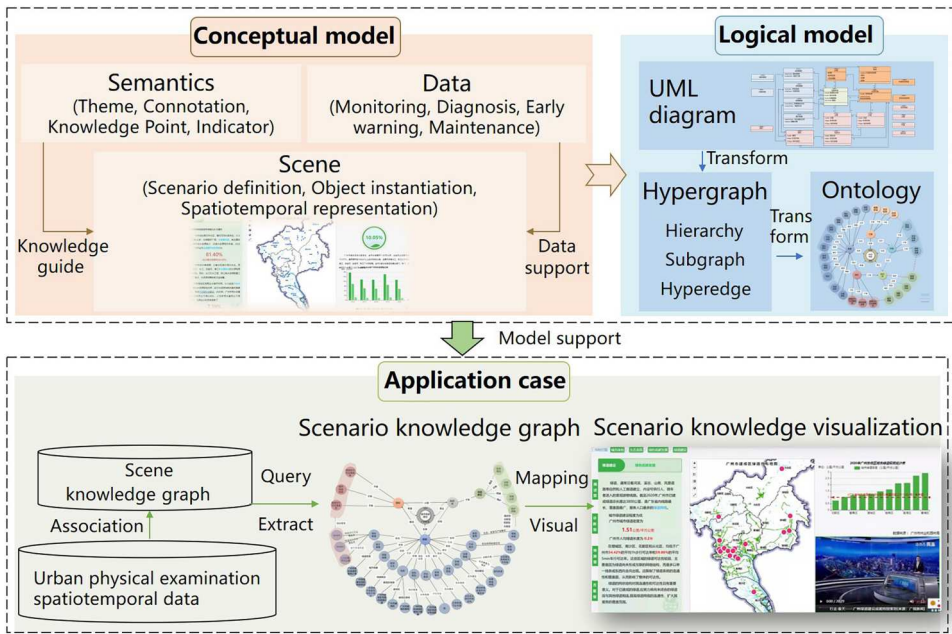


Figure 1. General framework for knowledge graph-based spatiotemporal data model for urban physical examination. This framework consists of a three-layer architecture of ‘semantics-data-scenario’, aiming to integrate multi-source spatiotemporal data and providing knowledge service for urban physical examination.

Secondly, this paper centers around application scenarios and constructs a logical model encompassing scenario identifiers, knowledge systems, and evidence data, and integrating logical relationships such as inclusion, inheritance, and extension. The logical model utilizes Unified Modeling Language (UML) diagrams to depict the logical structures of data, knowledge, and scenarios along with their spatiotemporal attributes. Concurrently, to enhance the expression of complex associations among spatiotemporal data, an ontology model based on a hypergraph structure is introduced. It represents higher-order relationships through hypernodes and hyperedges to optimize the expression of complex associations between spatiotemporal data.

Finally, this paper employs urban greenway construction as a case study for knowledge visualization and expression to demonstrate the application efficacy of the model. The case offers descriptive, diagnostic, predictive, and programmatic greenway knowledge services by gathering, integrating, and analyzing relevant data to comprehensively assess the state of urban greenway construction.

3. Conceptual modeling

Urban physical examination spatiotemporal data is multi-faceted, covering multiple types, time periods, and media. It encompasses natural background, population dynamics, environmental quality, etc. The forms include image, text, table, and video data, laying a solid foundation for in-depth urban physical examination analyses.

This paper differentiates between data and knowledge. Urban physical examination data refers to the spatiotemporal data collected from all aspects and stages of a city, such as basic attributes, functional structure, development status, and potential risks. It involves the city's monitoring, diagnosis, early warning, and maintenance stages (Wu et al. 2024). Urban physical examination knowledge is the result of processing and analyzing such data, including descriptive, diagnostic, predictive, and scenario-based knowledge about the city, as well as knowledge about its strengths, weaknesses, problems, suggestions, and countermeasures (Chen et al. 2020; 2021; Chen, Liu, and Tian 2022). The data and knowledge of urban physical examination are closely intertwined and mutually interactive. Data serves as the basis for knowledge, provides data support for scenario-based applications, while knowledge represents the distillation and application of data and guides scenario-based applications.

Firstly, this paper presents a triple conceptual model consisting of semantic, data, and scenario aspects to guide the development of a knowledge graph-based spatiotemporal data model. As shown in Figure 2, these layers correspond respectively to the knowledge, data, and application aspects of urban physical examination.

In detail, the semantic layer proposes a hierarchical structure for urban physical examination, including themes, connotations, knowledge points, and indicators. It constructs a comprehensive four-tier urban physical examination knowledge system, integrating both semantic and data entities into a unified framework. This aims to enhance the knowledge density and logical coherence of the data, strengthens the expression of data significance and interconnections, and provides a solid knowledge foundation for the scenario layer. The data layer organizes and refines relevant data knowledge according to the four key stages of urban physical examination: monitoring, diagnosis, warning, and maintenance. It aims to support the calculation of evaluation metrics, comprehensive diagnosis, causal analysis, and the improvement of irregularities in urban physical examination, thereby providing substantial data support for the scenario layer. The scenario layer instantiates the knowledge and data from the semantic and data layers, and realizes the visual representation of knowledge services in specific scenarios, facilitating the provision of urban physical examination knowledge services.

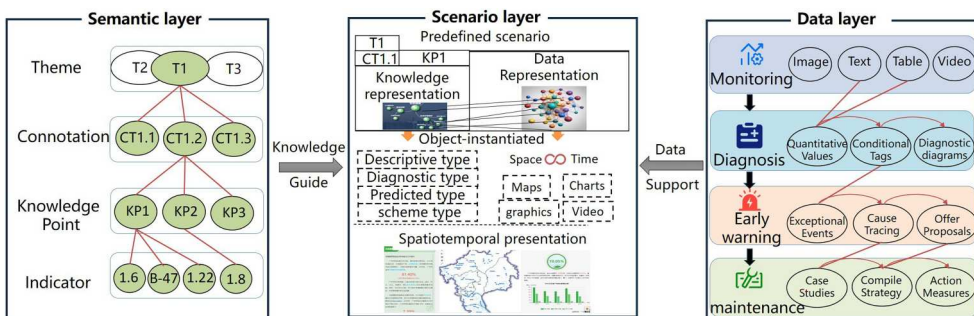


Figure 2. The conceptual model for knowledge graph-based spatiotemporal data model for urban physical examination. The model includes three layers: the semantic layer, the data layer, and the scenario layer, which are responsible for defining the knowledge of urban physical examination, managing spatiotemporal data, and providing scene-specific knowledge services, respectively.

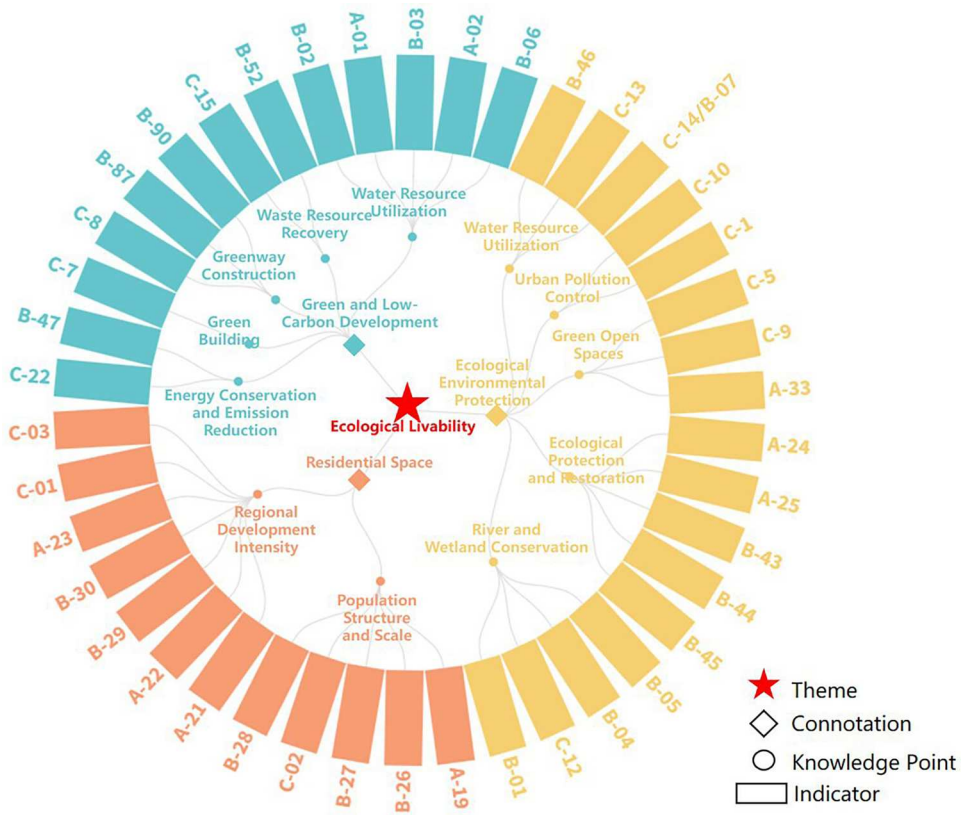


Figure 3. The four tiers of urban physical examination knowledge system: theme, connotation, knowledge point and indicator.

3.1. Semantic layer

This paper formulates a hierarchical urban physical examination knowledge system that consists of four levels: ‘theme-connotation-knowledge point-indicator’, as shown in Figure 3. The urban semantic and data entities are abstracted into a structured model, which facilitates the semantic modeling of urban physical examination concepts, their interrelations, and their spatiotemporal characteristics. This methodological improvement enhances data interpretability and relationship-expression capacity, ensuring alignment with the diverse knowledge demands of different urban physical examination scenarios. The formal representation of the semantic layer is described as follows.

$$S = (S_T, S_C, S_K, S_I) \quad (1)$$

In the equation, the term S with subscripted quotations (T, C, K, I) refers to a semantic set. S_T indicates the theme node, S_C represents the connotation node, S_K stands for the knowledge node, and S_I denotes the indicator node. The specific details of these four levels of nodes are presented below.

Table 1. Classification and definition samples of urban physical examination themes.

Urban physical examination theme	Theme definition
Ecological livability	This theme emphasizes the sustainability of the urban environment and the quality of life of its residents. It covers various aspects such as urban greening, air quality, and water resource management, aiming to create a green, healthy, and harmonious living environment.
Innovative vitality	This theme focuses on the city's capacity for innovation and economic dynamism. It includes elements such as technological innovation, industrial development, and business incubation, with the aim of promoting sustainable economic growth and enhancing the city's competitiveness.
Health comfort	This theme is concerned with the health and well-being of the city's residents. It covers areas such as medical services, leisure and entertainment, and community services to ensure that residents have access to high quality living services.
Convenient transportation	This theme looks at the city's transport systems and travel efficiency. It includes the optimization of transport networks, the development of public transport, and innovative traffic management, aiming to achieve fast, convenient, and safe urban travel.
Historical features	This theme is dedicated to the preservation and development of the city's historical and cultural characteristics. It pays attention to the city's historical sites, cultural activities, and architectural styles, with the intention of shaping the city's unique cultural identity and image.

3.1.1. Theme

The theme layer is the highest level of the knowledge system of urban physical examination, which provides a macro perspective and framework for urban physical examination. This paper takes Guangzhou city as an example, and selects the themes of Ecological Livability, innovation dynamism and health and comfort, etc., as delineated in Table 1. Based on the analysis of the relevant policies, plans and indicator index of urban physical examination, this paper explores the city's development goals and value orientation, and then adopts the form of ontology to define the classification, hierarchical structure, attributes and relationships of the themes of urban physical examination. The formal representation of the theme is presented as follows:

$$T = (T_H, T_A, T_R) \tag{2}$$

Where, T represents the set of themes. T_H refers to the theme's structure, T_A indicates the theme's attributes, and T_R signifies the theme's relationships. Specifically, T_H is defined as a tree data structure to represent the hierarchical relationships among themes. Each node within this structure represents a theme, and the edges between nodes indicate the parent-child relationships between themes. For example, the theme of Ecological Livability could contain sub-themes such as Ecological Environmental Protection and living Space. These sub-themes are manifested as nodes within T_H and are linked to the Ecological Livability theme node by edges. T_A is dictionary data in the form of key-value pairs, where the key is the name of the attribute related to the topic and the value is the descriptive information or quantitative value of the corresponding attribute, e.g. {'Eco-livability': {'Topic definition': ..., 'Resident satisfaction': 0.8}}. T_R represents the types of relationships or the intensity of associations among themes. For example, if there is a positive correlation between 'Environmental Livability' and 'Convenient Transport', the value of the corresponding element in T_R would be 0.7, indicating the degree of relationship between the two.

Table 2. Fundamental connotations of urban physical examination.

Connotations of urban physical examination	
Ecological and environmental protection	Network connectivity
Green and low-carbon development	Open development
living space	Public service provision
Green and efficient transportation	Innovation and development
Urban environmental maintenance	Urban vitality
Disaster prevention and mitigation	Care for vulnerable groups
Basic safety	Coordinated development
Comprehensive transportation services	Housing security
Historical and cultural identity	Urban-rural integration
Urban character	Community service guarantee

3.1.2. Connotation

Connotation is the second level, which provides in-depth understanding and analysis for the physical examination of the city, reflecting the connotation and relationship of the city's development. This paper analyses the points of concern according to the theme and indicators, and combines relevant literature and cases to condense the basic connotations of urban physical examination, as shown in Table 2. The connotations are related to each other. For example, the residential space connotation under the Ecological Livability theme is closely linked to the community service connotation under the Health and Comfort theme. The connotation is formally expressed as:

$$C_T = (T, C_j), \quad j = 1, 2, 3 \dots, n \quad (3)$$

Where, C_T denotes the connotations related to theme T , while C_j indicates the different connotative features subsumed under a particular theme, covering three dimensions: environment, society, and economy. Specifically, C_j assumes the form of dictionary data in key-value pairs. The key corresponds to the name of the connotation characteristic dimension, and the value embodies the dimensional information of the corresponding attribute, covering three dimensions of the urban ecological environment (such as green and low-carbon development paradigm, ecological environmental conservation), social livelihood (such as living space, community service guarantee), and economic development (such as open development, innovative development).

3.1.3. Knowledge point

The Knowledge Point represents the third level of the urban physical examination knowledge system, providing specific contents and dimensions for urban physical examinations, reflecting the current state of development and existing issues in cities. Based on the focal points of attention and knowledge analysis in the scene of urban physical examination, the classification, hierarchical structure, concepts, attributes, and relationships of knowledge points are defined in the ontological form, thereby delineating specific knowledge contents and dimensions. A part of urban physical examination knowledge points is presented in Table 3.

Urban physical examination knowledge points can be categorized according to their thematic domains. Taking the theme of Ecological Livability as an example, its knowledge points are further divided into three subcategories: Ecological and Environmental Protection, Green and Low-Carbon Development, and Living Space. Each subcategory includes specific knowledge points, forming a multi-layered structure. This hierarchical

Table 3. Knowledge points of urban physical examination.

Fundamental connotations	Knowledge points	Attributes	Knowledge content
Ecological and environmental protection	Ecological protection and restoration	Ecosystem service functions	Describes the various direct and indirect benefits that ecosystems provide to the city
		Ecological Risk Assessment	Conducts regular assessments of ecologically sensitive areas such as nature reserves, parks, and wetlands
		Restoration Techniques and Methods	Introduces current advanced ecological restoration technologies
		Policy and Regulatory Support	Outlines relevant policies and regulations that promote ecological protection and restoration
	River and wetland conservation	Ecosystem service functions	Describes the multiple ecosystem services provided by river and wetland areas
		Water resource management	Evaluates the role and impact of river and wetland ecosystems on urban water security and water quality
		Biodiversity conservation	Highlights the irreplaceable function of river and wetland ecosystems in maintaining biodiversity
		Public participation and education	Outlines measures to raise public awareness of the importance of protecting river and wetland areas
	Green open spaces	Functional Diversity	Describes the multiple functions of green open spaces
		Accessibility	Assesses the vehicular and pedestrian accessibility to green open spaces from residential areas
		Continuity	Evaluates the continuity and network of green open spaces
		Network	Outlines the social, economic, and cultural value of green open spaces as ecological corridors
	Water resource utilization	Resource sustainability	Evaluates the baseline conditions and long-term supply capacity of water resources
		Efficiency and Conservation	Assesses the efficiency of water resource use and promotes water-saving technologies and equipment
		Pollution Prevention and Control	Strengthens water quality monitoring and wastewater treatment
		Policy and Regulatory Support	Outlines relevant policies and regulations that promote water resource management
	Urban pollution control	Pollution Source Identification	Accurately identifies urban pollution sources, determining their type, distribution, and emission volumes
		Pollutant Characteristics	Analyzes the impact of different pollutants on human health and environmental quality
		Pollution Control Technologies	Introduces corresponding pollution control technologies and methods
		Policy and Regulatory Support	Outlines relevant policies and regulations that promote urban pollution control

classification method helps to systematic understand and analyze various aspects of the city. Each knowledge point represents a specific assessment dimension in the urban health evaluation. Attributes are detailed descriptions of knowledge points, offering four types of knowledge: descriptive, diagnostic, predictive, and solution. For example, the knowledge point of Ecological Protection and Restoration includes concepts such as biodiversity, ecosystem services, and the establishment and management of nature reserves, which form the basis for understanding and assessing urban ecological

health. Its attributes include ecosystem service functions, ecological risk assessment, restoration techniques and methods, and relevant policies. The relationships between knowledge points are categorized into four types: complementary, causal, correlational, and hierarchical. For instance, Green Open Spaces have a complementary relationship with Ecological Protection and Restoration, providing the spatial basis for the latter, which in turn ensures the sustainability and effectiveness of these services; Energy Conservation and Emission Reduction have a causal relationship with Ecological Protection and Restoration, with the implementation of the former promoting the effectiveness of the latter; Road Traffic has a correlational relationship with Efficient Commuting, with the improvement and expansion of the former effectively facilitating the latter; Urban Management has a hierarchical relationship with Urban Appearance, with the effectiveness of the former directly impacting the maintenance of the latter. The knowledge points are formally expressed as:

$$K_C = (C, K_j, K_A, K_R), \quad j = 1, 2, 3 \dots, n \quad (4)$$

In the formula, K_C refers to the knowledge points associated with the connotation C , K_j denotes a specific knowledge point within this connotation, expressed in textual form as a particular assessment dimension in the urban physical examination. K_A represents the attributes of this knowledge point and the corresponding detailed knowledge content, which requires a dictionary-structured data format. The key is the attribute name (textual in nature), and the value is the corresponding detailed knowledge content (comprising multiple data types such as text, numerical values, and lists. For example, under the ecological risk assessment attribute may contain specific risk numerical values and corresponding risk description texts), which is used to comprehensively describe the various characteristics of knowledge points and provide a basis for in-depth analysis. K_R represents the relationships between knowledge points, which represents the type of relationship between different knowledge points in the form of a text list, and is used to analyze the interactions and dependencies between knowledge points.

3.1.4. Indicator

The indicator layer represents the fourth level of the urban physical examination knowledge system, providing the assessment elements and their calculation methods for urban physical examination. Assessment elements are entities or things that reflect the assessment objects of the urban examination, such as residential neighborhoods, hospitals, higher education institutions, etc., and also include some actions in the physical world, such as ecological restoration, unauthorized use of land and sea, regional development, etc., as shown in Table 4. In Figure 3, under the knowledge point of Water Resource Utilization, indicators such as Water Resource Development and Utilization Rate, Water Consumption per Ten Thousand Yuan of GDP, etc., form the indicator model of water resource utilization. Table 5 presents examples of the definitions and calculation formulas of Water Resource Utilization indicators. Through the aggregation of indicators for urban physical examination and the construction of indicator models, quantitative evaluations for urban physical examinations are facilitated. The expression of indicators is as follows:

$$I_K = (K, I_j, I_A, I_R), \quad j = 1, 2, 3 \dots, n \quad (5)$$

Table 4. Assessment elements of urban physical examination.

Elements of MOHURD	Elements of MONR
Regional development	High-standard agricultural land
Air quality	Basic agricultural land
Carbon dioxide emissions	Land use efficiency
Population density	Water resources
Green building	River and lake ecosystems
Parks and green spaces	Urban vitality
Greenways	Care for vulnerable groups
Construction land	Coordinated development
Ecological corridors	Housing security
Surface water quality	Urban-rural integration
Reclaimed water	Community service guarantee
Residential solid waste	Water use efficiency
Residential communities	Energy infrastructure
Housing security	Energy use efficiency
Cultural buildings	Wetland conservation
Urban waterlogging	Groundwater resources
Traffic accidents	Reclaimed water utilization
Emergency shelters	Water consumption
Hospitals	Emergency refuges
Parking spaces	Fire and rescue services

In the formula, I_K refers to the indicators related to the knowledge point K , I_j refers to a specific indicator within this knowledge point, presented in textual form and indicating the name of a specific quantifiable assessment element. I_A embodies the attributes of this indicator, which is a constitute composite data, containing indicator elements (mainly text-based) and the corresponding indicator-related attributes (such as the formula, calculation method and description of the data source, units, and other relevant text or numerical type information), which are employed to precisely define the specific connotations and computational rules of the indicators. And I_R represents the relationships between indicators, which helps to analyze the synergistic effects between indicators and other relevant circumstances in textual form.

In the comprehensive analysis of the urban physical examinations of the city, Equation 1 is first used to construct a complete semantic set and build a knowledge framework covering all aspects of the city, which integrates different levels of knowledge at the macro level and provides a basis for subsequent in-depth analysis. Focusing next on the

Table 5. Definitions and calculation formulas for water resource utilization indicators.

Indicator name	Definition	Calculation formula	Remarks(unit)
Total water resources	The total amount of water resources available for use in a city annually.	$= (\text{total precipitation} - \text{evaporation})$	m^3
Water resource utilization rate	The proportion of actual water usage to the total available resources	$= (\text{total population of the city} / \text{total water resources})$	%
Per capita water usage	The annual available water resources per person	$= (\text{total population of the city} / \text{total water resources})$	$\text{m}^3 / \text{person}$
Reclaimed water reuse rate	The proportion of reclaimed water usage in the total water consumption	$= (\text{actual water usage} / \text{reclaimed water usage})$	%
Industrial water efficiency	The amount of water consumed per unit of industrial output in industrial production	$= (\text{total industrial output value} / \text{total industrial water usage})$	$\text{m}^3 / \text{ten thousand yuan}$

*Notes:

(a) This table provides basic indicators and methods for assessing the status of urban water resource utilization.

(b) The calculation of indicators is based on data from the city's statistical yearbook and the water affairs bureau.

(c) The calculation formulas in the table are for reference only and could be adjusted according to the actual applications.

theme level, the structure, attributes of different themes and their interrelationships are analyzed through Equation 2 to understand the main direction of urban development and the synergies or constraints between themes, such as judging the strength of association between the theme of Ecological and Livable and the theme of Convenient Transport on the overall layout of the city. Equation 3 is then used to delve into the connotation level under the theme. to analyze the specific connotation characteristics of each theme, and to interpret the essence of the city's development in terms of the theme from multiple dimensions, such as the environment, society, economy. With the help of Equation 4, we could further refine to the knowledge point level. Based on the knowledge points corresponding to the connotations, their attributes and relationships, to explore the current situation and problems of the city in specific assessment dimensions and the interactions between the knowledge points. Finally, Equation 5 is implemented at the level of indicators, through specific indicators and their calculation methods and relationships, to achieve a quantitative evaluation of all aspects of the physical examination of the city, to provide a precise data base for evaluating the effectiveness of urban development and diagnosing problems, and, based on the results of these different levels of analysis, to make a comprehensive judgement of the overall state of health of the city, to identify existing problems, and to make targeted development proposals.

3.2. Data layer

The data layer is based on the concretization of concepts and relationships of the urban vital signs. These are the ontologies and instances within the semantic layer. The data layer is used for data collection, analysis, evaluation, and optimization. It includes spatiotemporal data for the monitoring, diagnosis, early warning, and maintenance stages of urban physical examination. The data layer provides the meaning, attributes, and relationships of spatiotemporal data for each stage of the urban physical examination, thereby facilitating the understanding and exchange of data. Among these, the data source and spatiotemporal attributes are crucial semantic content for spatiotemporal data. The data source describes the acquisition modalities and origin channels of urban physical examination data, including government departments, enterprise institutions, social media, and Internet of Things devices. The spatiotemporal attributes of the data describe the spatiotemporal dimensions and features of urban physical examination data, including elements such as data timestamp, spatial scope, spatial resolution, and temporal granularity. The formalized expression of the data layer is presented as follows:

$$E = (D_T^p, D_S^p, D_A^p, D_R^p) \quad (6)$$

In the formula, E represents the collection of data elements, p denotes the stage depth of the data element in the urban physical examination process, D_T indicates the data type, D_S signifies the data source, D_A represents the data attributes, and D_R denotes the relationships between the data.

3.2.1. Monitoring phase

Spatiotemporal data will be collected and integrated from multiple data sources during the monitoring phase of the physical examination of the city, which cover indicators

Table 6. Data content at four stages of the data layer.

Stage	Data type	Data content	Data source	Data attributes	Relationship type
Monitoring	image	urban surface, architecture, transportation, greening, etc., such as remote sensing images, traffic photos, greening photos, etc.	acquired through satellite remote sensing, drone photography, ground cameras, etc.	image resolution, color, contrast, etc.	spatial location, temporal sequence, content similarity, and other relationships of images
	text	public opinion, sentiment, policy, etc., in the city	obtained through social media, news media, government announcements, etc.	publication time, title, organization, etc.	relationships among texts such as theme ownership, authorship, citation, etc.
	table	population, economy, environment, etc., in the form of tables, such as census data, GDP statistics, air quality, etc.	acquired through government statistics, corporate reports, expert research, etc.	number of rows, columns, field names, etc.	relationships expressed through primary keys, foreign keys, joins, etc.
	video	activities, events, phenomena, etc., in the city, such as traffic congestion, public safety, urban culture, etc.	utilizing live streaming, surveillance cameras, mobile devices, etc.	video duration, resolution, frame rate, etc.	time sequence, content similarity, and other relationships of videos
Diagnosis	numerical values	values of various indicators of urban physical examination, such as development level, speed, direction, etc.	calculated through mathematical formulas, statistical methods, machine learning, etc.	size, range, precision, etc., of numerical values	relationship operations such as addition, subtraction, multiplication, division, comparison, sorting, etc.
	conditional labels	tags attached to various indicators of urban physical examination, such as development types, stages, features, etc.	attached through classification, clustering, annotation, etc.	tag name, category, quantity, etc.	relationship judgment such as containment, parallelism, mutual exclusivity, etc.
	diagnostic graphics	graphics of various indicators of urban physical examination, such as development trends, distribution, relationships, etc.	generated through visualization, graph theory, network analysis, etc.	graphic type, color, size, etc.	relationship analysis such as similarity, comparison, mapping, etc.
Early Warning	exceptional events	detection of exceptional events in urban development	analyzed through anomaly detection, identification, classification, etc.	event name, time, location, etc.	relationship analysis such as causality, sequence, concurrency, etc.
	cause tracing	identification of causes of abnormal events in urban development	analyzed through association analysis, causal analysis, impact analysis, etc.	cause name, weight, confidence level, etc.	relationship analysis such as correlation, independence, conflict, etc.

(Continued)

Table 6. Continued.

Stage	Data type	Data content	Data source	Data attributes	Relationship type
Maintenance	offer proposals	proposals for exceptional events and causes	referenced and supplemented with data from the maintenance stage	proposal name, content, effect, etc.	relationship analysis such as applicability, optimization, supplementation, etc.
	case studies	successful or failed cases of comprehensive governance for abnormal indicators in urban development, such as typical cases, lessons learned, reference value, etc.	acquired through case collection, case analysis, etc.	case name, time, location, etc.	similarity, comparison, reference, etc., among cases
	compile strategy	comprehensive governance strategies for abnormal indicators in urban development, such as adjustment plans, improvement measures, optimization goals, etc.	formulated through optimization models, decision support, multi-objective planning, etc.	strategy name, content, effect, etc.	prioritization, coordination, supplementation, etc., among strategies
	action measures	comprehensive governance action reports for abnormal indicators in urban development, such as execution status, effect evaluation, feedback, etc.	generated through execution supervision, effect assessment, feedback adjustment, etc.	action name, content, executing unit, etc.	execution measures, summary, updates, etc., among actions

under multiple perspectives, such as spatial planning and human settlements, and provide a solid foundation for the assessment of the physical examination of the city. As shown in Table 6, these indicator data cover different types, formats and qualities of data sources, such as satellite remote sensing, Internet of Things, social media, official statistics, etc., and they are characterized by multi-scale and multi-granularity in both space and time.

Processing spatiotemporal data from different sources involves categorizing and handling each type accordingly. Specifically, for remote sensing data, necessary preprocessing steps such as geometric correction, and radiometric calibration are performed and then standardized to the GeoTIFF format and the WGS84 coordinate system to meet the needs of subsequent analysis (Sharma et al. 2024). For social media data, text cleaning and standardization is performed to unify the encoding to UTF-8 to ensure data quality (Schmitt et al. 2019). The processing of government statistical data includes format conversion, missing value handling, unit conversion and standardization of indicators to ensure consistency, completeness and comparability. For online video data, key elements such as events, times, and locations are extracted and transformed into structured attribute data to ensure accuracy and completeness (Khan et al. 2024).

The attributes of these data include type, format, and quality, etc. The integration relationships among them ensure the consistency and reliability of the data. At the same time attributes reveal the granularity, scope and distribution of the data with spatial and temporal transformation relationships. For example, the data are divided according to spatial (national to province, city, district, street) and temporal (year to month, day, hour, minute, second) scales. Or divide it from different perspectives, such as spatial planning, human settlement environment, and other perspectives. Attributes clarify the themes, indicators and criteria of the data, as well as the analytical relationships between the data, facilitating flexible analyses based on different concerns and evaluation criteria.

3.2.2. Diagnosis phase

The diagnostic phase of the physical examination of the city uses spatiotemporal analysis and mathematical and statistical methods to measure and evaluate the level of indicators for various aspects of urban development. The data in this phase are based on the indicator results of the data from the monitoring phase and involve three data types: numerical, labeled and graphical.

Diagnostic numerical data are dynamically updated and reflect real-time changes in urban development. The attributes include the diagnostic name, the target value, and the relationships used for data updating, such as data sources, computational procedures, and evaluation metrics. Diagnostic labels, on the other hand, reveal urban conditions and regional characteristics, showing the horizontal and vertical hierarchical differentiation of the data. The attributes of the labels detail the hierarchical structure and categorical characteristics of the data, along with the data aggregation operations through relationships such as summarization, averaging, and sorting. Then the diagnostic graphics present the city's strengths, problems, and potentials in the form of intuitive charts. Its attributes describe the type and evaluation criteria of the graphs, as well as the complex influencing, facilitating and constraining relationships between the data. Ultimately, through diagnosis, it is possible to more accurately grasp the dynamic trend of urban development and identify the level of development and characteristics of each region, thereby providing scientific decision support for the warning and decision-making stages.

3.2.3. Early warning phase

The early warning stage is based on the resultant data of the diagnostic stage and uses data mining and reasoning methods to further analyze the abnormal indicators present in the city physical examination. The data in this stage are the early warning results of the physical examination of the city and contain three data types: events, causes and recommendations.

Event data reveal key issues in urban development, such as diseases, pitfalls and opportunities, through the analysis of anomalous indicators, with attributes that include the risk and value of the event, as well as relationships between the data, including early warning, intervention, and improvement measures.

Cause data, on the other hand, identifies the root causes of anomalous events through correlation and causal analyses, with attributes describing the characteristics and predictions of the causes, as well as the relationships between the data, such as triggering, propagation, and blocking.

Suggested data make appropriate suggestions for abnormal events and causes, with hierarchical interaction and stage adjustment characteristics, while attributes relate to the stages, assessment and optimization of suggestions, as well as feedback and adjustment relationships between data, such as input, output and update. By further investigating data mining and predictive reasoning, the warning phase could identify and respond to potential risks in urban development in a timely manner, and provide targeted recommendations to provide a scientific basis for strategies in the maintenance phase.

3.2.4. Maintenance phase

The maintenance phase is based on the resultant data from the early warning phase to optimize and manage the anomalous indicators and to form a comprehensive management strategy and corresponding action plan to promote urban development. The types of data in this phase include case studies, strategies and actions. The case studies include both successful and unsuccessful comprehensive governance cases, which have been collected and analyzed in the context of urban development. They serve to highlight typical cases, lessons learned, and the significance of these cases for future reference. The strategy represents the comprehensive urban governance strategies developed for abnormal indicators through optimization models, decision support, and multi-objective planning, such as urban improvement measures and optimization targets. The action measures reflect the comprehensive governance action reports for urban development. These are generated through the processes of execution supervision, effectiveness evaluation, and feedback adjustment. They include details on the city's implementation status, performance assessment, and feedback received.

These maintenance outcome data not only reflect the city's physiological condition in aspects such as ecology, economy, and society but also demonstrate the level of development in terms of efficiency, quality, and equity. They support and facilitate the city's achievement of sustainable and high-quality development goals. The attributes encompass aspects such as maintenance data types, improvement measures, and potential stimulation, as well as supervisory and evaluative relationships like execution, inspection, feedback, improvement, and stimulation, providing a solid data foundation for urban management and decision-making.

3.3. Scenario layer

The scenario layer is responsible for the organization of knowledge and data from the semantic and data layers into specific application instances. This is done in accordance with the diverse needs of urban physical examination monitoring, diagnostics, warnings and maintenance. It provides spatiotemporal representation of the scenarios and objects pertaining to urban vital signs. The objective is to develop a knowledge service for urban health scenarios, providing scenario-specific knowledge services aligned with diverse application domains, including data services, analysis services, prediction services, evaluation services, and optimization services for urban physical examination.

In conclusion, the scenario knowledge is visually expressed in applications through the utilization of appropriate visualization techniques, in accordance with the specific forms of application, such as charts, maps, reports, and narratives derived from urban

physical examination. The above visual representation facilitates the intuitive and vivid presentation of scenario knowledge, thereby enhancing the effectiveness of information communication and the accuracy of decision-making. The formal expression of the scenario layer is as follows:

$$\text{Scene} = (\text{Scene}_T, S, E, R_{\text{Scene}}) \quad (7)$$

In the formula, Scene represents the set of application scenarios, Scene_T denotes the type of application scenario, S signifies the semantic set, E indicates the set of data elements, and R_{Scene} represents the scenario relationships.

This section begins with three key aspects of scenario construction: pre-definition scenario, object-instantiated, and spatiotemporal presentation of scenarios. Utilizing technologies such as data mining, knowledge graphs, and visualization, it extracts and integrates data and knowledge related to scenarios, providing users with scenario knowledge services.

3.3.1. Pre-defined scenario

This paper defines the various types of scenarios and their respective spatiotemporal scopes through the pre-definition of scenarios, thereby facilitating the accommodation of a multitude of diverse application scenarios. For example, the spatiotemporal scope of a scenario can be selected and adjusted according to the specific analytical requirements of the user, allowing for the consideration of different levels of granularity and perspectives. A scenario template is a customizable and easily expandable approach to structuring scenarios. It encompasses semantic and data objects pertinent to the scenario, such as domain, objectives, content, and form, as well as their interrelationships, including causality, influence, and optimization.

The term ‘semantic objects’ is used to describe the various semantic concepts that are embedded within the urban physical examination knowledge graph. These include themes, connotations, and knowledge points, which possess specific semantic meanings and attribute characteristics. Such descriptions can be employed to delineate the multifarious aspects and levels of knowledge pertaining to urban physical examination. In contrast, data objects refer to the data elements that quantify and assess each stage and level within the urban physical examination knowledge graph. These include indicators and data, which are characterized by their numerical values and units of measurement. To illustrate, scenario templates make direct use of themes, connotations, knowledge points, indicators and data from the semantic layer for the purpose of pre-defining knowledge content. They employ a variety of data forms, including textual, cartographic, tabular, and pictorial, to express the logical and numerical connections between knowledge and data. This allows for the presentation of the status and characteristics of a specific aspect or level of urban physical examination.

3.3.2. Object-integrated

This paper employs the semantic and data objects integrated by the scenario knowledge template to retrieve and extract relevant entities from the semantic and data layers. The above semantic and data objects are instantiated as entities within the urban physical examination knowledge graph, with edges representing relationships such as belonging,

containment, and influence. This serves to articulate the logical structure and internal connections of urban physical examination knowledge.

The process of object instantiation is a pivotal step in the construction of the urban physical examination knowledge graph, enabling a detailed and nuanced representation of the city's health status. By meticulously mapping out the relationships between various entities, the knowledge graph provides a comprehensive framework that facilitates the analysis of complex urban health issues.

3.3.3. Presentation of scenario spatiotemporal data

This paper maps and transforms entities of spatiotemporal data from urban physical examination knowledge graphs, such as text, maps, charts, images, and videos, into spatiotemporal expressions of scenarios. It generates scenario objects, attributes, and relationships that describe and display the application scenarios of urban physical examination knowledge services. The term 'scenario objects' is used to describe elements representing various aspects and levels of urban physical examination knowledge services. These may include knowledge systems and evidence data, which possess both spatiotemporal and formal characteristics. The visual features of urban physical examination knowledge presentation, including color, shape, size, and other visual elements, are referred to as scenario attributes. These attributes possess visual and effect characteristics and can be used to represent urban physical examination. The relationships between scenario objects express the spatiotemporal structure and external manifestations of urban physical examination scenarios, including positions, dynamics, and statics in visual layouts.

Based on the generated scenario objects, attributes, and relationships, the paper establishes rules and methods for mapping and transforming spatiotemporal presentations. Utilizing technologies such as data mining, knowledge graphs, and visualization, it realizes the spatiotemporal expression of urban physical examination scenarios, exemplified by the greenway network map on maps, greenway indicator analysis in charts, greenway appearance display in images, and greenway benefit demonstration in videos.

4. Logical model and ontology

This paper focuses on application scenarios and employs Unified Modeling Language (UML) diagrams to articulate the logical model of these scenarios, as shown in [Figure 4](#). It describes the logical structure and relationships of data, knowledge, and scenarios, as well as their spatiotemporal attributes. The urban physical examination knowledge service encompasses multiple application sets, including monitoring, diagnostics, warnings, maintenance, evaluation, and knowledge. Each application set has its own expression of application scenarios, which is composed of attributes such as scenario identifiers, knowledge systems, and evidence data, and possesses relationships of inclusion, inheritance, and extension.

To facilitate the expression and management of the spatiotemporal dimensions of data, knowledge, and scenarios, this paper introduces three elements into each UML class diagram: spatiotemporal tags, spatiotemporal indexes, and spatiotemporal versions. Spatiotemporal tags are labels used to describe the spatiotemporal characteristics of data, knowledge, and scenarios. They encompass features including time, location, scope, and

the ontology design, the hypergraph consists of hypernodes and hyperedges. Each node and edge are tagged with spatiotemporal labels, indexes, and version information. The model uses hyperedges to represent high-order relationships, enhancing the expression and management of data across spatiotemporal dimensions and complex associations (Zhang et al. 2025). Each node and edge is accompanied by spatiotemporal tags, spatiotemporal indexes, and spatiotemporal versions. Moreover, a hypergraph is introduced to represent high-order relationships, thereby better realizing the expression and management of the spatiotemporal dimensions and complex associations of data. Among them, a subgraph is a part of the graph model, representing a group of nodes and edges of data, such as a subgraph of a theme, a subgraph of knowledge, etc. A subgraph can contain other subgraphs, forming a hierarchical graph structure, such as a monitoring subgraph, a diagnosis subgraph, a warning subgraph, a maintenance subgraph, etc. The formal expression of the graph model is as follows:

$$G_E = (G_H, G_{sub}, R_H) \quad (8)$$

In the formula, G_E represents the graph model of data element E , G_H denotes the

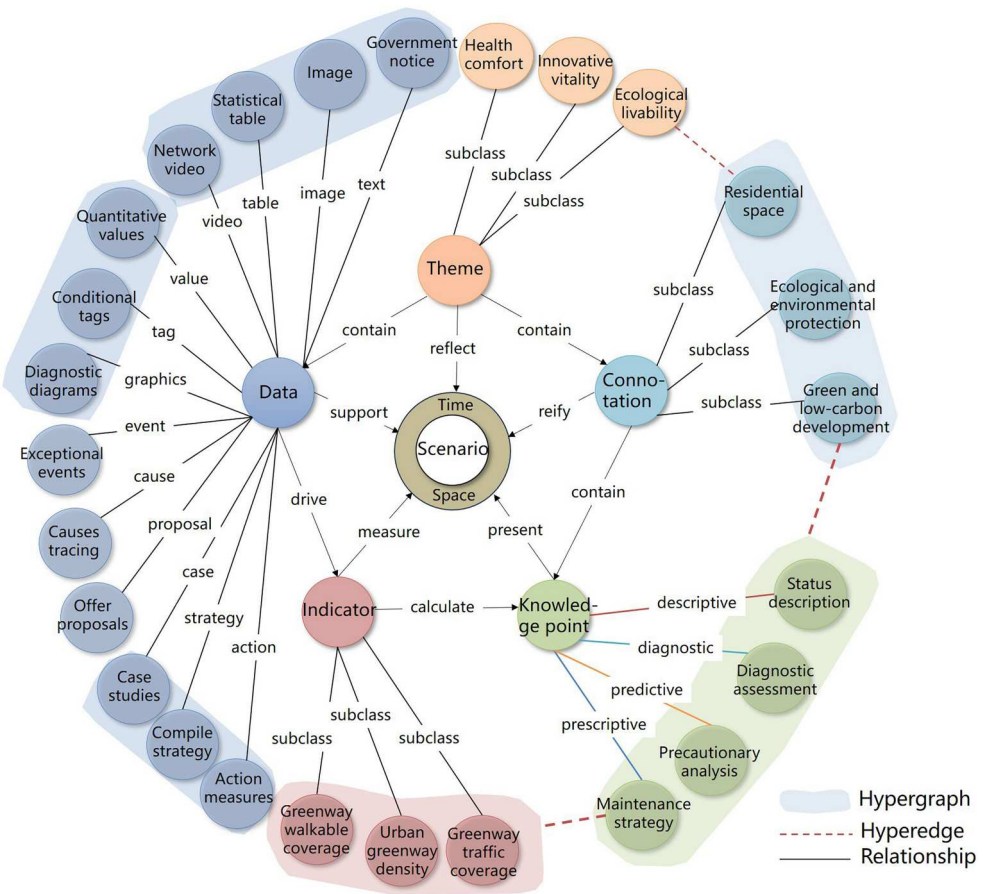


Figure 5. The ontology of knowledge graph-based spatiotemporal data model for urban physical examination. It uses hypergraphs to simplify the expression of complex relationships.

hierarchical structure of data elements in the urban health process, reflecting spatiotemporal scale, health assessment stage depth, and differentiation characteristics; G_{sub} indicates a data subgraph composed of a set of nodes and edges, embodying semantic features; R_H signifies the higher-order relationships within the data hypergraph.

$$G_H = (S_{\text{scale}}, T_{\text{scale}}, P, F) \quad (9)$$

Here, S_{scale} represents the spatial scale, T_{scale} represents the temporal scale, P denotes the depth of the urban physical examination stage, and F indicates the differentiation characteristics of the urban physical examination. Among them, S_{scale} and T_{scale} are typically numerical data or enumeration data with specific encodings. They are utilized to clarify the spatial and temporal scopes or hierarchies involved in urban physical examinations. For instance, the spatial scale can be represented by geographical area codes (such as urban district codes, block codes, etc.) or specific numerical values of spatial ranges (such as area size values). The time scale can be in the form of timestamps or time-period codes (such as years, quarters, etc.). P and F are either numerical values denoting depth levels or text-based encodings with specific semantics. They are employed to describe the stage at which the urban physical examination is conducted (such as the monitoring stage, the diagnostic analysis stage, etc.) as well as the specific characteristics of the physical examination (such as by functional areas, by knowledge types, etc.).

$$G_{\text{sub}} = (G_D, g_A) \quad (10)$$

In the formula, G_D represents the subgraph dimensional features, and g_A denotes the attributes of the subgraph. Here, G_D is a collection data type that combines numerical and text types, used to represent the set of characteristic identifiers of the subgraph in different dimensions (such as theme dimension, knowledge dimension, indicator dimension, etc.). It is a dictionary-type data. The keys are various attribute names of the subgraph (such as the name of the subgraph, creation time, associated data object type, etc.), and the values are the specific contents of the corresponding attributes (text, numerical values or other data types), which are used to describe the characteristics of the subgraph in detail. g_A is a dictionary data. The keys are various attribute names of the subgraph (such as the name of the subgraph, creation time, associated data object type, etc.), and the values are the specific contents of the corresponding attributes (text, numerical values or other data types), which are used to describe the characteristics of the subgraph in detail.

$$R_H = (R_S, R_T, R_P, R_F, R_D) \quad (11)$$

Here, R_S signifies the spatial relationships between hierarchical structures, R_T represents the temporal relationships, R_P denotes the connection relationships between physical examination stages, R_F indicates the differentiation relationships between vital sign features, and R_D represents the feature relationships between subgraph dimensions. Among them, the relationships are associative dictionary-type data structures, in which the elements or values represent attribute information such as the type and intensity of the relationship.

Specifically, hypernodes can represent data objects, semantic objects, and scenario objects in urban physical examination, such as indicators, results, events, themes, connotations, knowledge points, monitoring objects, diagnostic objects, warning objects, etc. Hyperedges can represent data relationships, knowledge relationships, and application relationships in urban physical examination, such as calculation, evaluation, association, influence, invocation, etc. In addition, there are also some relationships between hypernodes and hyperedges, such as hierarchy, combination, decomposition, parallelism, exclusivity, conflict, etc., which can express the structure and logic of data, knowledge, and applications.

In the mechanisms for maintaining the timeliness and ensuring the quality of knowledge graph, consistency verification during the update process is a critical technical link. When a knowledge graph is updated, each newly added relationship and entity is systematically checked against the existing knowledge corpus. For example, it checks whether new relationships conflict with pre-existing semantic constraints or inference rules, and whether the attributes of new entities conform to ontology definitions. If conflicts are detected, adjustments are made either manually or through specific algorithms to ensure the logical coherence and semantic accuracy of the knowledge graph. By systematically validating new information against established ontological frameworks and logical axioms, knowledge graph can dynamically incorporate new knowledge while maintaining structural integrity.

5. Application case

To validate the effectiveness and practicality of the knowledge graph-based spatiotemporal data model for urban physical examination proposed in this paper, the construction of urban greenway in Guangzhou City was selected as an application case for in-depth study. A prototype system was developed on the Python platform, using the Django framework and the Neo4j graph database to implement the specific application of the model. As an essential component of the urban ecological network, greenways not only enhance the ecological quality of the city but also improve the living environment for residents, making them an ideal subject for this experimental validation (Semeraro et al. 2021; Van Oijstaeijen, Van Passel, and Cools 2020). Taking the construction scenario of urban greenways in Guangzhou as an example, through the collection, integration, and analysis of relevant data related to greenway construction, the system identifies the problems and risks in greenway construction and provides scientific decision support.

5.1. Spatiotemporal data processing and storage

The processing and storage of spatiotemporal data for the urban greenway construction in Guangzhou includes requirements analysis, data collection and collation, node connection and index establishment, as well as hierarchical storage.

Firstly, to effectively integrate and store spatiotemporal data related to urban greenway construction, the system begins with a needs analysis to clarify the requirements for consolidating data and knowledge on urban greenways. Urban greenways, as a type of nature-based urban green infrastructure, primarily rely on non-motorized transport

and walking. They aim to meet residents' needs for leisure, fitness, and tourism. The spatiotemporal data involved cover multiple aspects, including ecology, landscape, transportation, and society. These data should reflect the impact of greenways on residents' lives, such as greenway density, greenway walking accessibility rate, and greenway driving accessibility rate.

Next, according to the data type requirements of each node in Formulas (1–7), spatiotemporal data related to greenway construction is collected from different channels and collated into data forms suitable for scene nodes, theme nodes, connotation nodes, knowledge nodes, indicator nodes, etc. For example, spatial data of greenways is extracted from the Geographic Information System, and after calculating indicator data such as greenway density, it is stored in the indicator node and a correlation relationship with the theme node Ecological Livability is established, indicating that this indicator is an important basis for measuring the greenway construction situation under the theme of Ecological Livability. Then, using the spatiotemporal data model of urban physical examination based on knowledge graph, connections and indexes between nodes are established to organize and express the relevant data and knowledge of greenway construction. These scattered node data are integrated into a scene set 'Scene', which provides a basic data framework for the subsequent analysis and application of the relevant data and knowledge of greenway construction, as shown in [Figure 6](#). Adopting a hierarchical storage architecture, the system stores the semantic layer data of the knowledge graph, the raw data of the data layer, and the application data of the scenario layer separately. A close association among these spatiotemporal data is established through unique identifiers and index mechanisms.

5.1.1. Semantic layer

In the semantic layer, various semantic entities within the spatiotemporal data of urban greenway construction are stored as nodes in the knowledge graph. Each node is assigned a unique identifier within the neo4j graph database and stores the attribute information related to the entity. For instance, an urban greenway density indicator node may possess attributes such as the indicator name (urban greenway density), the calculation method (e.g. a specific formula), and the unit (kilometers per square kilometer). Semantic relationships between nodes are represented by edges. For example, an urban greenway density indicator node and a diagnostic assessment knowledge point node are connected by an edge labeled used for calculation, and the edge can also have attributes, such as the weight of the relationship (indicating the strength of the relationship, e.g. the importance weight of this indicator in certain calculations).

5.1.2. Data layer

The original data in the data layer is stored in appropriate file systems or key-value stores based on its data type and source. Concurrently, data nodes are created in the knowledge graph, and reference links pointing to the storage locations of the original data are established in neo4j. This enables accurate retrieval and processing of the original data when required. For example, for knowledge points, geographic information data, and online video resources related to greenway construction nodes, a common indexing key (such as the unique identifier of the greenway and the geographical area code) is established in neo4j to quickly associate and retrieve the original data.

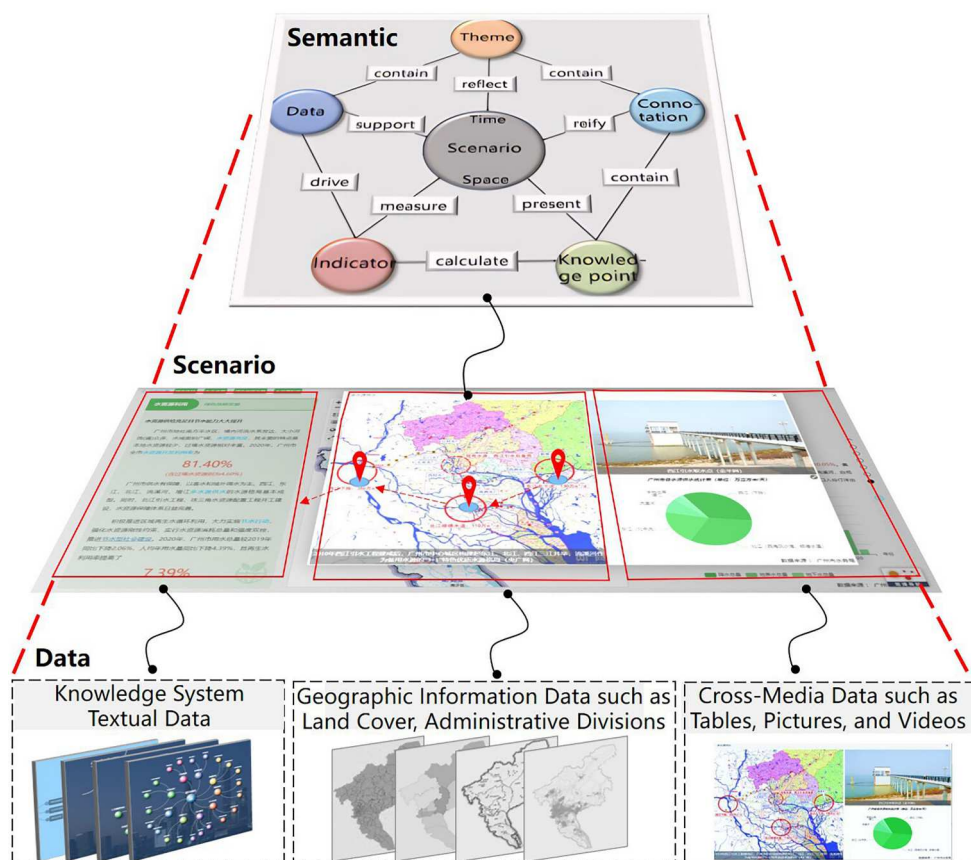


Figure 6. Diagram of spatiotemporal data processing and storage in knowledge graph-based spatiotemporal data model for urban physical examination.

5.1.3. Scenario layer

For the scenario layer data, in combination with application requirements, the configuration information related to the scene (such as the scene type and spatiotemporal range predefined in the scene) and the user interaction data (such as the operation records of the user in the scene visualization) are stored in the form of attribute tables in either the file system or a relational database. Simultaneously, scene nodes are created in neo4j, and the data and knowledge in the attribute tables are associated with these scene nodes. For example, in the case of a Guangzhou urban greenway construction scene node, the initial setting parameters of the scene (such as the theme, region, and time span) are retrieved through the associated attribute table, and the semantic knowledge and data required for the application scene are obtained through these nodes.

5.1.4. Relationships and indexing in neo4j

Neo4j stores a variety of relationships. These relationships encompass spatial relationships (such as adjacency, containment, intersection, etc.), temporal relationships (such as sequential order, contemporaneity, etc.), causal relationships (such as the impact of greenway construction on wetlands, the promotion of the integration of urban

ecology, recreation, and industry by greenway construction, etc.), and semantic relationships (such as hierarchical relationships between concepts, synonymous relationships, etc.). The relationship attribute records information including the type of relationship, the identifiers of the starting and ending nodes, and the weight of the relationship (indicating the strength of the relationship).

To enhance the efficiency of data querying and retrieval, multiple indexing mechanisms are established. For example, a spatial index based on geographical location is implemented to expedite the querying of urban physical examination data within specific regions. A temporal index based on timestamps is utilized for retrieving data in chronological order. Indexes based on the key entity types and attributes of the knowledge graph, such as crucial concepts like themes, connotations, knowledge points, and indicators, are also created to facilitate knowledge querying.

5.2. Spatiotemporal data analysis based on knowledge graph

The spatiotemporal data analysis of urban greenway construction in Guangzhou based on knowledge graph involves ontology and knowledge graph construction, and analysis of greenway construction achievements.

5.2.1. Ontology and knowledge graph of greenway construction

The ontology of the urban greenway construction knowledge graph is designed in accordance with Formulas (8–11). Firstly, by clarifying the hierarchical structure information of the spatiotemporal data of urban greenway construction in terms of spatiotemporal scale (in 2020, Guangzhou/City District), urban physical examination stage (diagnosis, warning, and maintenance stage), and differentiation characteristics (city/district regional differentiation), G_H is constructed. Then, by using the urban greenway density indicator to analyze and evaluate the overall situation of urban greenway construction, and analyzing and warning the connectivity of the urban greenway network through the walkability and drivability rates of greenways, and further providing maintenance strategies in combination with the differentiation characteristics of urban greenways, the ‘Greenway Diagnosis- Warning-Maintenance Subgraph’ is formed, and G_{sub} is constructed. Moreover, the connection relationships between G_H and G_{sub} are analyzed. For example, indicator values, statistical tables, and their indicator graphic data drive the calculation of indicators such as urban greenway density, marked as ‘Data Driven-Indicator Calculation’; the results of these indicator calculations obtain the knowledge content of greenway diagnosis, warning, and maintenance, marked as ‘Indicator Calculation-Diagnosis, Warning, and Maintenance Knowledge Generation’; the obtained knowledge points further require evidence factual data such as online videos, government announcements, and typical cases for support, marked as ‘Diagnosis, Warning, and Maintenance Knowledge Generation-Data Support’. These high-order relationships in the hypergraph comprehensively describing the urban greenway construction data are used to construct R_H . Finally, G_H , G_{sub} , and R_H are combined into G_E , forming a knowledge graph G_E of urban greenway construction, thereby comprehensively presenting the achievements and characteristics of greenway construction in Guangzhou, as shown in [Figure 7](#). Among them, the node types can be divided into knowledge point nodes (such as the current situation, diagnostic evaluation, warning analysis, and maintenance

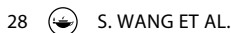


Figure 7. The Guangzhou greenway construction scenario knowledge graph. Key nodes (scenario, knowledge point, indicator, and data) and their relationships are highlighted.

strategy of greenway construction), indicator nodes (such as the walkability coverage rate of greenways, urban greenway density, and drivability coverage rate of greenways), and data nodes (such as specific geographical entities such as greenways, built-up areas, and road networks; indicator graphics such as statistical charts and distribution maps; governance strategies such as planning and construction, ecological restoration, and functional value, etc.). The edge types represent various relationships between nodes, such as subclass relationships (such as the urban greenway density indicator node is a subclass of the indicator node), calculation relationships (such as the urban greenway density indicator node is used to calculate the diagnostic evaluation knowledge point node of greenway construction), and expression relationships (such as the urban greenway construction scene node expresses descriptive, diagnostic, predictive, and schematic knowledge points of greenway construction).

As depicted in Figure 8, a spatiotemporal data analysis comprehensive analysis of the greenway construction outcomes in Guangzhou based on knowledge graph is presented as follows.

5.2.2. Theme

The greenway construction in Guangzhou is centered around the theme of Ecological Livability, with the overarching objective of enhancing the ecological environment

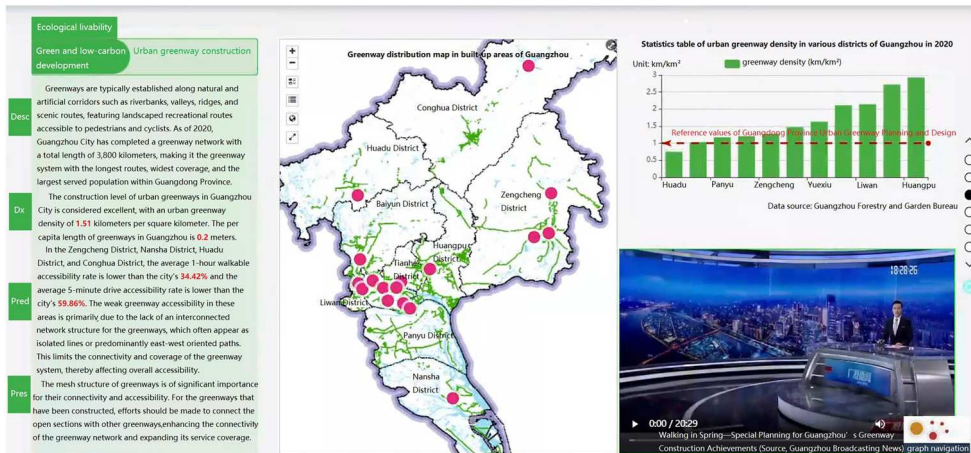


Figure 8. Spatiotemporal data analysis of Guangzhou greenway construction. It shows the status quo of greenway construction, development trend and the relationship between various elements.

quality of the city and improving the living standards of its residents. This theme serves as the guiding principle for all aspects of greenway planning and development, aiming to create a harmonious coexistence between urban development and natural ecosystems.

5.2.3. Connotation

The connotation of Green and Low-Carbon Development is well manifested in greenway construction. The construction of greenways has effectively promoted the connectivity of urban green spaces, providing increased opportunities for residents to engage in green travel. This, in turn, contributes to the reduction of carbon emissions and propels the city towards a healthy and sustainable green development path. By integrating greenways into the urban fabric, the city can enhance its ecological resilience and reduce its ecological footprint.

5.2.4. Indicator

The urban greenway density in Guangzhou is measured at 1.51 kilometers per square kilometer, with a per capita greenway length of 0.2 meters. The average 1-hour walking accessibility coverage rate of greenways is 34.42%, and the average 5-minute driving accessibility coverage rate is 59.86%. However, it is notable that in the Zengcheng District, Nansha District, Huadu District, and Conghua District, these indicators fall below the average levels of the entire city. This disparity indicates a relatively significant variation in the distribution of greenways within the urban area, suggesting potential areas for improvement and optimization.

5.2.5. Knowledge point

The construction of greenways in Guangzhou is comprehensively analyzed from various knowledge dimensions such as the current situation, assessment, early warning, and coping strategies, as described below:

Current Status: The greenway construction in Guangzhou has achieved remarkable progress in terms of the current. By 2020, the total length of completed greenways in

the city had reached 3,800 kilometers, forming a relatively comprehensive greenway network. This extensive network provides a valuable recreational and ecological resource for the city's residents.

Diagnostic Evaluation: The overall evaluation of the greenway construction effect is rated as excellent. This assessment is based on a comprehensive analysis of various factors, including the physical infrastructure, ecological functions, and social benefits of the greenways.

Warning Analysis: By identifying spatial differences in regional greenways, the warning analysis has identified several potential problems. In particular, the walking and cycling coverage of greenways is relatively low in some areas. Projections indicate that with the continued urban expansion and population migration to areas such as Zengcheng District and Nansha District, if the current pace of greenway construction remains unchanged over the next five years, the accessibility coverage rates in these regions are expected to decline further. The estimated average 1-hour walk accessibility coverage rate may fall to 25%, and the average 5-minute drive accessibility coverage rate may fall to 45%. This projected decline is mainly due to fact that the rate of urban expansion far exceeds the rate of greenway construction, and existing greenways often exist as isolated segments or are predominantly oriented in an east-west direction, limiting the connectivity and coverage of the greenway system and adversely affecting the user experience.

Maintenance Strategy: In response to these challenges, a series of maintenance strategies are proposed.

- (1) **Planning recommendations:** In the process of developing new urban areas and renovating old districts, the planning and construction of greenways should be synchronized, and special land should be reserved for greenway construction. Taking advantage of the natural geographical resources in these regions, such as the water systems in Zengcheng District and the mountain ranges in Conghua District, new greenway routes should be strategically planned and integrated with the existing network.
- (2) **Optimize connectivity:** Special attention should be given to urban fringe areas and densely populated regions to enhance the overall accessibility of the greenway system. For existing greenway segments that remain unconnected, efforts should be made to link them with other greenways to improve the connectivity of the network and expand its service coverage. At the breakpoints and areas with weak connectivity in the greenway network, the construction or renovation of connectivity facilities is essential. For example, where greenways are interrupted by rivers, motorways or other obstacles, bridges, tunnels and footbridges can be built to provide seamless connections.
- (3) **Financial support:** It is recommended that the government set up special greenway construction funds for new construction, maintenance and upgrading of greenways. At the same time, introduce policies to encourage social capital to participate in greenway construction, such as granting tax incentives and naming rights. Collect residents' needs and suggestions for greenway construction through community research and public feedback, and improve greenway usage and satisfaction.



Figure 9. Visual expression of scene knowledge in urban greenway construction helps users to quickly understand relevant knowledge.

5.3. Scene knowledge visualization

The system provides urban greenway construction knowledge services by leveraging the urban greenway construction knowledge graph to visually demonstrate the characteristics and achievements of the urban greenway construction. The urban greenway construction scene knowledge visualization is presented in Figure 9.

5.3.1. Integration of spatiotemporal data

The spatiotemporal data model of urban physical examination based on knowledge graph integrates spatiotemporal data from multiple data sources, such as the built-up area scope, road network, and greenway distribution in Geographic Information System (GIS), population data from statistical departments, action reports, governance strategies, and typical case data from government departments, as well as multimodal data like online videos. Through the hierarchical storage architecture and unique identifier and indexing mechanisms within the model, these data are organically integrated to form a complete basic data of urban greenway scenes.

5.3.2. Structured-knowledge mining

Furthermore, the model defines the hierarchical structure information of urban greenway construction spatiotemporal data in terms of spatiotemporal scales (e.g. 2020, Guangzhou City/District), physical examination stages (diagnosis, early warning, and maintenance stages), and differentiation characteristics (city/district regional differentiation). This assists planners in understanding the progress of greenway construction in different time periods and the distribution differences of greenways in different regions, as depicted in Figure 9. Additionally, multi-dimensional knowledge of urban greenways is mined, covering aspects from the connotation level of 'Green and Low-carbon Development' to various coverage indicators at the indicator level and



Figure 10. Problem diagnosis and maintenance measures in urban greenway construction scenarios. At the macro-city scale, the focus is on diagnosing the overall layout and planning issues of urban greenways. At the meso-region scale, the emphasis is on diagnosing the connectivity, balance, and building issues of greenways between different regions. At the micro-street scale, the diagnosis targets the actual usage experience and facility details of greenways at the street level.

maintenance strategies at the knowledge point level. The correlation analysis of this multi-dimensional knowledge provides a comprehensive decision-making basis for urban planning.

5.3.3. Interactive scene visualization

The spatiotemporal data of the urban greenway construction scene is visualized through interactive visualization technology, providing users with intuitive and vivid urban physical examination knowledge services. As shown in Figure 10, the distribution map of greenways in built-up areas of Guangzhou can clearly show the location and orientation of greenways in the city. This enables urban planners and decision-makers to clearly understand the current spatial layout of urban greenways. Besides, through street-view images of greenways at macro-level city, meso-level region, and micro-level street spatial scales, the regional differences in greenway construction are intuitively analyzed. Meanwhile, combined with greenway construction planning measures and supporting images, the utilization and ecological effects of the greenway, such as residents' leisure activities on the greenway and the growth and change of vegetation around the greenway, are dynamically displayed. Thus, the function and value of the greenway are more comprehensively demonstrated.

In summary, the knowledge graph-based spatiotemporal data model for urban physical examination deeply analyzes the current situation and existing problems of greenway construction in Guangzhou, and puts forward targeted decision-making suggestions. In the data layer, by integrating multi-source spatiotemporal data, it provides a solid foundation for diagnosing problems in greenway construction. In the semantic layer, through constructing a knowledge system of 'theme-connotation-knowledge point-indicator', it

dives into the problems in greenway construction and offers decision-making knowledge. In the scenario layer, through visualization, it clearly presents the current situation, problems and optimization strategies of greenway construction, providing a scientific basis for urban planners and decision-makers. This model transcends the limitations of conventional models and facilitates a systematic understanding of urban green space construction through its comprehensive, systematic and in-depth organization and expression capabilities. The application of this model marks an important transformation of urban physical examination work from individual data analysis and indicator diagnosis to systematic knowledge services.

6. Discussion

It was found that the successful development for knowledge graph-based spatiotemporal data model for urban physical examination is largely contingent upon numerous favorable conditions. These include the adoption of a ‘semantics-data-scenario’ three-layer architecture for modeling, securing innovative interdisciplinary collaboration, and achieving deep user engagement and feedback.

6.1. Applicability and localization of the model

In this study, a knowledge graph-based spatiotemporal data model for urban physical examination with applicability and localization was successfully constructed through structured semantic modeling, associative data integration, and scenario-oriented knowledge representation. The model is not only capable of addressing the diversity and complexity of urban physical examination data but also allows for customization according to the unique needs of individual cities, accommodating various scales and localized urban physical examination.

Initially, to ensure that the knowledge graph comprehensively reflects all dimensions of urban physical examination while avoiding excessive complexity, the model adopts a four-tier structure: ‘theme-connotation-knowledge point-indicator’. This structure divides the knowledge into logical layers, covering all necessary aspects of urban physical examination and ensuring the model’s representativeness. Secondly, to maintain the integrity of knowledge points while reducing redundant connections, the model uses a hypergraph structure to simplify grouped or clustered knowledge. By representing closely related knowledge as hypernodes and hyperedges, the model eliminates nodes of redundant and weakly connected knowledge. This reduces model complexity and improves computational efficiency. Thirdly, to provide customized knowledge based on different application needs, the model converts abstract data and knowledge into intuitive charts, maps, and reports through scenario-oriented knowledge representation. This enhances the model’s interpretability.

In the case of Guangzhou’s greenway construction, the model identified two key issues: insufficient connectivity and uneven coverage of greenways. To address the connectivity issue, it is recommended to prioritize greenway layout in new urban planning. Natural geographical resources, such as water systems in Zengcheng District and mountain ranges in Conghua District, should be fully utilized to plan new greenway routes. When greenways are interrupted by rivers, highways, or other obstacles, bridge,

tunnel, and pedestrian overpass projects should be actively implemented to ensure the continuity of greenways. To address the issue of uneven coverage, community participation is recommended to gather residents' needs and suggestions for greenway construction. This will improve greenway usage and satisfaction. Additionally, a regular evaluation mechanism for greenway maintenance should be established, and social capital should be involved in maintenance efforts.

In the above-mentioned case, the spatiotemporal data comes from multiple channels, including remote sensing imagery, social media feedback, government statistics, and online videos. While these data sources provided rich information, they also had limitations and potential biases. First, remote sensing data had limited spatial resolution, leading to inaccuracies in measuring greenways precisely. Government statistics, though authoritative, were often not updated promptly, resulting in a lag between the data and the actual state of greenway construction. Social media and online video data could offer real-time feedback but lacked representativeness. For example, the voices of the elderly or those from remote areas might be overlooked. Second, the absence of detailed data and the presence of data biases made it difficult to conduct thorough quantitative analyses. Third, when assessing the model's applicability across different cities and regions, we found that urban physical examination goals and priorities are closely tied to a city's geographical location, economic structure, and socio-cultural background. Therefore, the model's customizable design allows urban planners and decision-makers to select the most suitable evaluation indicators and knowledge nodes based on local conditions, enabling localized application of the model.

6.2. Comparison with existing spatiotemporal data models

In recent years, research on spatiotemporal data models has continued to evolve, including relational database models (Capris et al. 2022), spatiotemporal data warehouse models (Reddy 2021), and spatiotemporal object models (Santhanavanich et al. 2022). Each model has its own strengths and limitations in terms of spatiotemporal data processing capabilities, applicable scenarios, and constraints. As shown in Table 7, a comparative analysis of the strengths, limitations, and applicable scenarios of existing models is provided.

Traditional relational databases, such as MySQL and PostgreSQL, excel at handling structured data but struggle with complex spatiotemporal relationships. In contrast, spatiotemporal data warehouse models, like the star and snowflake schemas, are advantageous for data analysis and reporting but lack flexibility in updates and struggle to meet the needs of real-time data analysis. Spatiotemporal object models, such as those based on OGC (Open Geospatial Consortium) standards, perform well in spatial data processing but fall short in semantic association and knowledge mining capabilities.

The model proposed in this paper has several advantages. First, it can effectively integrate heterogeneous data from different sources and supports multidimensional spatiotemporal data analysis. Second, through the structure of knowledge graphs, it can capture complex entity relationships and semantic information, enhancing data interpretability and application value. Finally, intuitive scene visualization helps decision-makers better understand the data and promote collaboration and communication across departments. However, the model also has some limitations. For example, its

Table 7. Comparison of spatiotemporal data models.

Data model	Strengths	Limitations	Applicable scenarios
Relational database model	Efficient for processing structured data and supporting complex queries	Struggles with spatiotemporal data and complex relationships	Traditional data storage and management
Spatiotemporal Data Warehouse Model	Suitable for data analysis and reporting, supports historical data queries	Inflexible for updates, poor real-time performance	Data analysis and decision support
Spatiotemporal Object Model	Strong spatial data processing capabilities, compliant with geographic information standards	Insufficient semantic association, limited knowledge mining capabilities	Spatial data analysis and applications
Knowledge Graph-Based Spatiotemporal Data Model	Integrates multi-source heterogeneous spatiotemporal data, supports complex relationships and semantic mining, strong dynamic updating capabilities	Dependent on the quality of spatiotemporal data, high complexity in construction	Urban physical examination, smart city management

effectiveness depends on the quality of the input spatiotemporal data, and building a knowledge graph requires significant technical expertise and resource investment.

6.3. Innovative interdisciplinary collaboration

The research and development for knowledge graph-based spatiotemporal data model for urban physical examination is a complex and multifaceted process that inherently involves a confluence of disciplines, including geographic information science, data science, and urban planning. The successful execution of this study unequivocally emphasizes the crucial and indispensable nature of close-knit collaboration and seamless knowledge sharing within interdisciplinary teams (Frantzeskaki et al. 2020). In addition, the establishment of effective mechanisms for interdisciplinary cooperation is essential to taking research in this field to new heights.

In the construction and practice application of the urban physical examination knowledge graph, a diversified collaborative platform has been established, leveraging the Guangzhou workstation of academicians as a catalyst. This platform is designed to foster synergistic cooperation among the realms of urban planning, geographic information science, data science, and artificial intelligence. By congregating experts and scholars from diverse fields, it effectively dismantles traditional disciplinary boundaries and barriers, thereby enabling a concerted effort to jointly grapple with the key issues and formidable challenges endemic to urban physical examination. To illustrate, urban planning experts proffer a macroscopic perspective and furnish specific case studies, geographic information scientists are entrusted with the task of data collection and processing, data scientists deploy advanced analytical techniques to unearth latent patterns and trends, and artificial intelligence specialists embark on the exploration of intelligent solutions with the aim of augmenting the model's predictive and decision-making capabilities. Through the implementation of astute team management and communication strategies, such as the establishment of a common language framework, the regular convening of interdisciplinary workshops, and the creation of platforms dedicated to the exchange of shared databases and research findings, the seamless flow of information and the harmonious integration of knowledge among team members are facilitated.

Consequently, this augments the overall work efficiency and efficacy of the team, thereby laying a solid foundation for the successful realization of the research objectives.

6.4. Deep user engagement and feedback

User engagement and feedback play a crucial role in the development and application process for the knowledge graph-based spatiotemporal data model for urban physical examination (Metzler et al. 2003). This intricate process mandates not only the forging of a close collaborative alliance between technical personnel and urban physical examination practitioners but also the extensive assimilation of perspectives and suggestions proffered by residents and other stakeholders. The profundity and expanse of user engagement exert a direct and profound impact on the model's practical utility and user contentment.

In the initial instance, direct user participation endows the model with invaluable firsthand data and real-world application scenarios. Urban physical examination professionals, drawing on their expertise and experience, can identify potential strengths and weaknesses of the model in the analysis of urban physical examination. Residents, as direct experiencers of urban life, furnish feedback that is instrumental in enabling the model to more precisely capture and mirror the exigencies and predilections of the populace. Other stakeholders, such as environmental organizations and business enterprises, are likewise capable of contributing sagacious insights into the model from their respective vantage points. This diverse feedback is indispensable for improving the model's functionality and optimizing its interface. In return, the urban physical examination knowledge services based on this model offer several benefits. For urban planners, it provides a scientific basis for planning and helps develop strategies that better meet the city's needs. For city decision-makers, it offers comprehensive and accurate data and knowledge support to enhance digital urban management. For the public, it provides a clear picture of urban development, increasing their sense of participation and identification with urban growth and environmental protection. Ultimately, this promotes sustainable social development.

In addition, to ensure the effective utilization of user feedback, the implementation of perpetual and robust mechanisms for user engagement is of the utmost exigency, as this serves as the linchpin for the model's long-term and sustainable development. To galvanize active feedback and user participation in the iterative amelioration process of the model, a repertoire of measures has been instituted, including regular user surveys, feedback systems for knowledge services, and public forums. Through such initiatives, users are not relegated to the status of passive recipients of the model; rather, they are actively enlisted as stakeholders in its evolutionary journey. This approach not only expedites the continuous enhancement of the model's capabilities but also engenders a sense of user ownership and satisfaction, thereby augmenting the model's effectiveness and societal acclaim. A model that garners widespread acceptance and active utilization by its users is more predisposed to catalyze the democratization and transparency of urban planning and management, thereby amplifying its value and conferring greater benefits upon society.

7. Conclusions

Facing the demands for integrating multi-source heterogeneous spatiotemporal data and providing customized knowledge services in urban physical examination, this paper proposes a knowledge graph-based spatiotemporal data model for urban physical examination. The model not only enriches the theoretical framework of spatiotemporal data organization and management in urban physical examination but also clarifies the transition path from theoretical models to technical applications. It innovatively adopts a three-layer architecture of ‘semantics-data-scenario’. From the perspectives of the semantic layer, data layer, and scenario layer, it constructs an urban physical examination knowledge system, manages the entire process of spatiotemporal data in urban physical examination, and provides scenario-oriented spatiotemporal representation. First, addressing the issue that current urban physical examination spatiotemporal data models focus on new indicator design and conceptual discussions without fully considering the application needs of urban physical examination, the ‘semantics-data-scenario’ conceptual model is proposed. It integrates knowledge from the semantic layer and data from the data layer based on application needs and presents them through the scenario layer in an intuitive and interactive visual manner to provide customized knowledge services to users. Second, to solve the problem of traditional UML diagrams that use complex nesting and associations to represent and process complex relationships, resulting in high complexity and poor readability, the ‘semantics-data-scenario’ logical model is proposed. It uses a logical model combining UML diagrams and hypergraph ontology, representing higher-order relationships as hyperedges to simplify the model structure and improve the accuracy of multidimensional relationship expression. Finally, taking the greenway construction in Guangzhou as a case study, the model demonstrates an innovative application mode of urban physical examination knowledge service.

The model transforms the traditional urban physical examination approach, which relies on experience and single indicators. It comprehensively organizes and expresses urban data and knowledge through knowledge graph, analyzing and mining the intrinsic connections between greenway construction and urban ecology, transportation, and population at multiple levels (themes, connotations, knowledge points, and indicators) and granularities (macro-level cities, meso-level regions, and micro-level streets). This helps understand urban greenways from multiple dimensions. Additionally, the model’s scenario-oriented spatiotemporal representation presents abstract knowledge and data in intuitive forms such as greenway distribution maps and indicator analysis charts. This clearly shows the current situation, issues, and development trends of greenway construction, making it easier for urban planners, decision-makers, and the public to understand and promoting multi-department collaboration and public participation.

Subsequent research can delve into performance testing and exploration at three levels: data, semantics, and application scenarios, with the aim of further enhancing the model’s universality and practicality. Specifically, the research can focus on extracting richer and more profound information and knowledge from data, thereby strengthening the model’s analytical depth and intellectual content. Additionally, the scope of the model’s application should be expanded to explore its potential role in key areas such as urban disease prediction and environmental risk assessment. Furthermore, we will

place emphasis on how to integrate this model with other technologies in smart cities to construct more efficient and sustainable urban management strategies.

Acknowledgements

We thank the editors and all the anonymous reviewers for their insightful comments and suggestions. We thank the Guangzhou Municipal Forestry and Landscaping Bureau for providing valuable data on greenway construction and the Guangzhou Alpha Software Information Technology Co., Ltd. for technical support in spatiotemporal data processing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research is funded by the National Key Research and Development Program of China [grant number 2022YFB390420501].

Data available statement

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials.

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